

Crown and bridge
All-ceramic crown
Preparation
(Jacket crown)

LECTURE 7

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Jacket (all-ceramic) crown

The most esthetically pleasing fixed restoration. Because, there is no metal understructure to block light transmission, it can resemble natural tooth in term of color and translucency than can any other restoration.



Jacket (all-ceramic) crown

It is that type of crown that is formed from tooth colored materials, it is mainly used as a single crown in anterior quadrants of the mouth. It is the weakest type of crowns because tooth colored materials are weak and brittle.

1- Acrylic jacket crown.

2- Porcelain jacket crown.



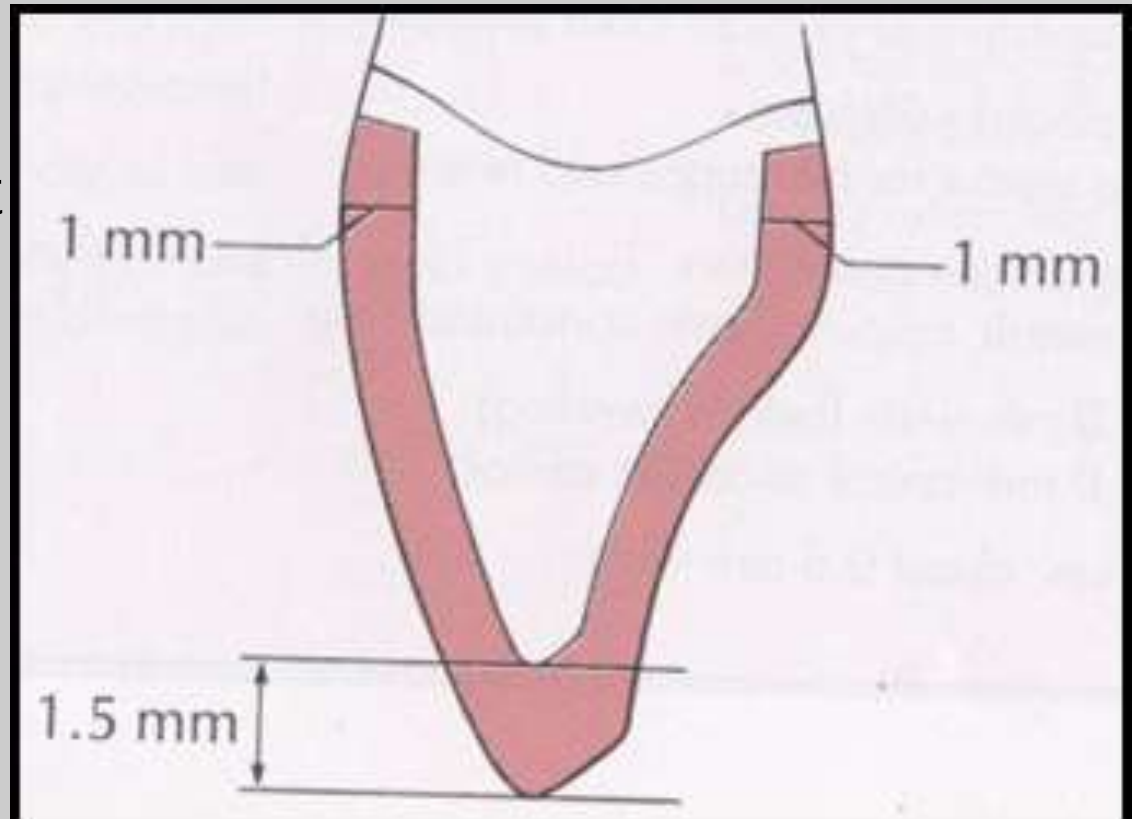
It is **not a conservative** type of crown preparation because shoulder finishing line must be prepared all around and excessive tooth structure will be removed to provide enough space for porcelain material to allow for proper **shaping and shade adjustment** of the crown and to **increase strength** of porcelain and to resist the fracture by increasing its thickness.



Preparation requirements:

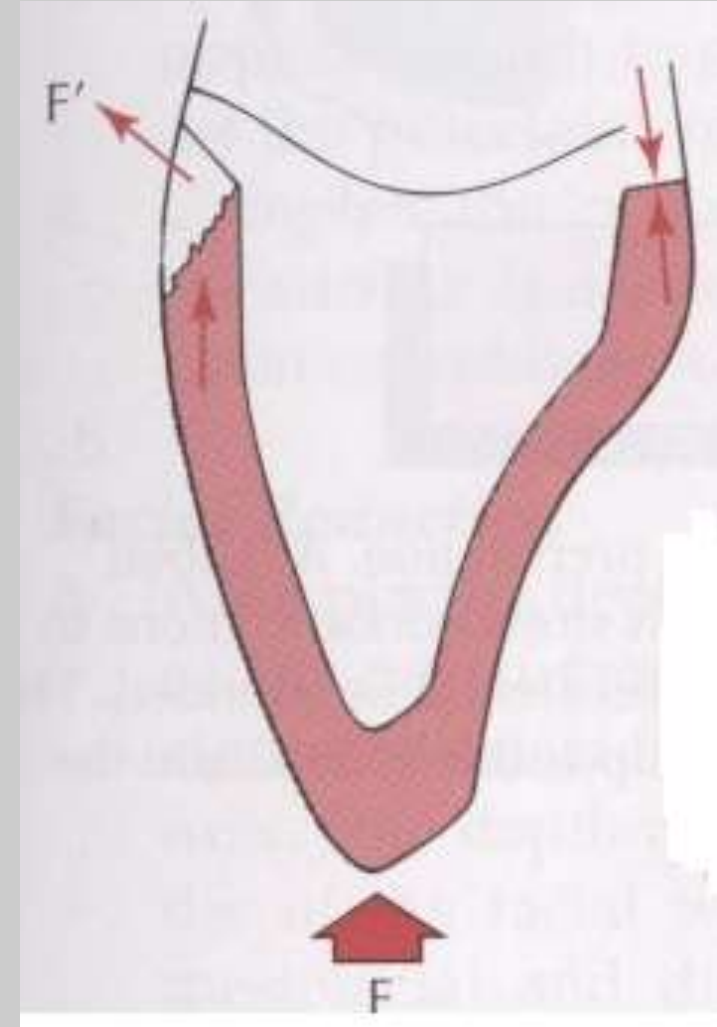
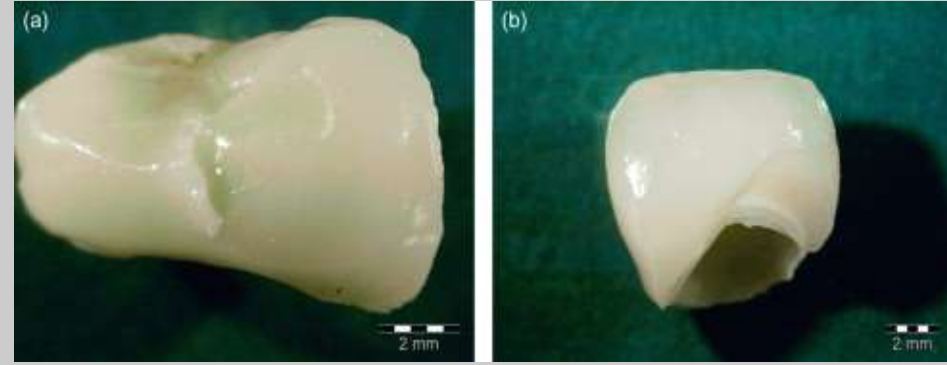
1- A shoulder of uniform width 1 mm is used as gingival finishing line to provide a flat seat to resist forces directed from the incisal direction. **A 90-degree shoulder is needed to prevent unfavorable distribution of stresses.**

2- The preparation must be as long as possible to give support to porcelain.



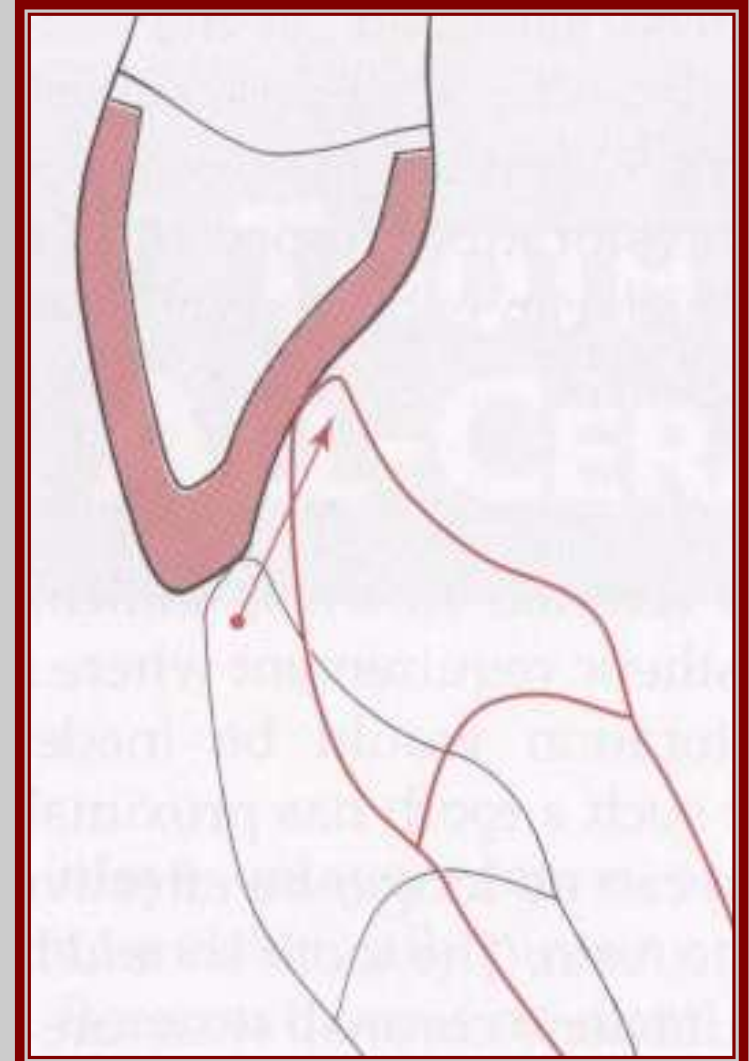
Preparations should be left as long as possible and avoid inclined type of shoulder.

An over shortened preparation will create stress concentrations in the labiogingival area of the crown, which will lead to a characteristic ‘half –moon’ fracture in the labiogingival area of the restoration.



3-The incisal edge is prepared **flat** and placed at a slight inclination toward the **linguogingival direction** to meet forces on the incisal edge of the lower opposite tooth crown to prevent shearing incisal part of the ceramic crown.

While for lower anterior teeth, preparation of incisal edge must be prepared with labial Inclination to give more Thickness for it to avoid its fracture.



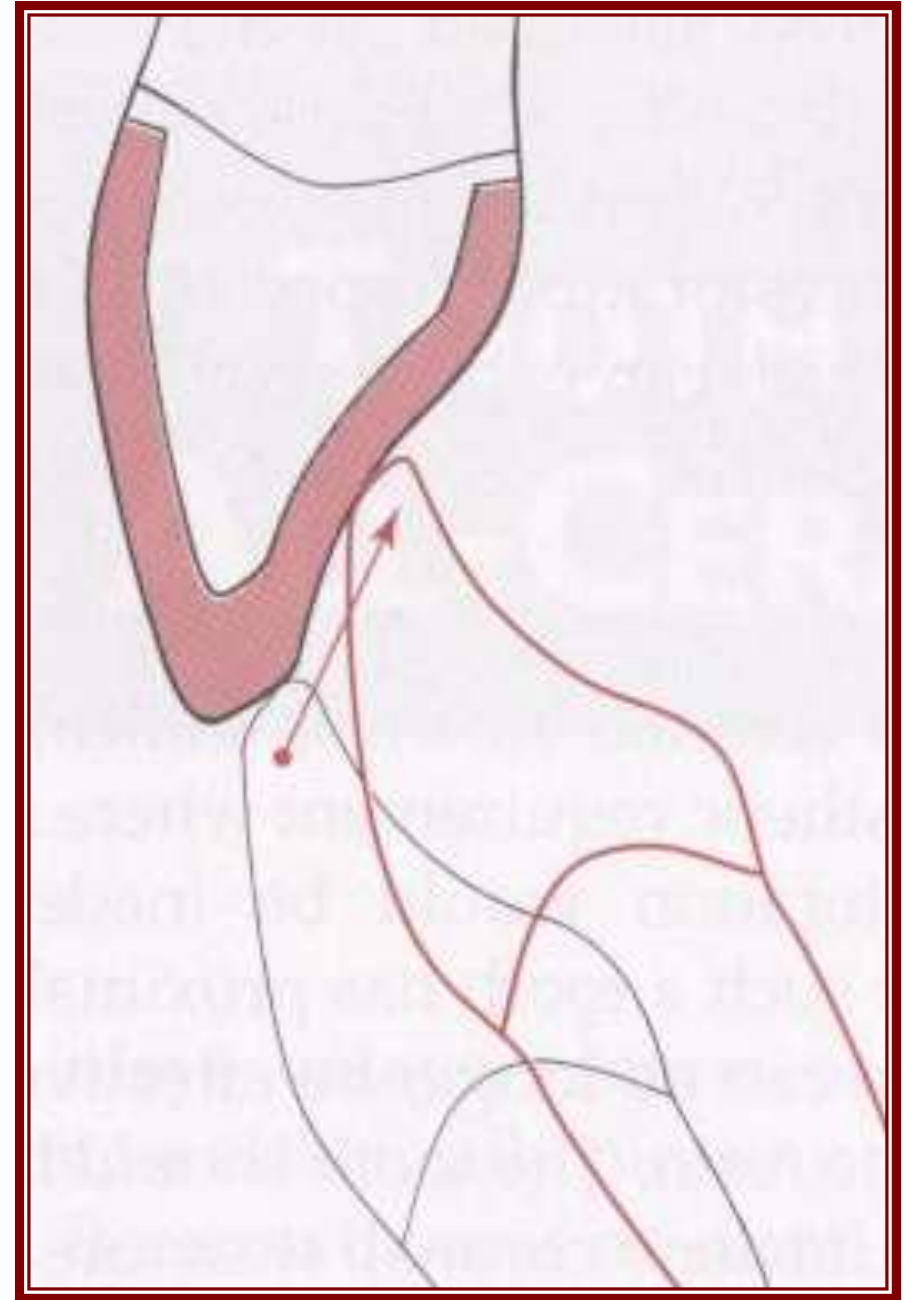
4-All sharp angles of the preparation should be rounded to reduce the danger of fracture caused by stress concentration.

5- It must be avoided from teeth with edge to edge relation.

6- The occlusion on all-ceramic crown is **critical** for avoiding fracture in centric contacts. Centric Contacts are best Confined to the middle third of lingual surface.



Because of the relative weakness of the restoration, the occlusal load should be favorably distributed. Generally this means that centric contact must be in an area where the porcelain is supported by tooth structure (e.g., in the middle third of the lingual wall).



Advantages of all-ceramic crown:-

- 1- Superior esthetics, excellent translucency.**
- 2- Good tissue response.**
- 3- Labially more conservative than metal – ceramic crown (lack of reinforcement by a metal substrate).**
- 4- The appearance of the complete restoration can be influenced and modified by selecting different colors of luting agents.**

Disadvantages:-

1- Reduced strength due to absence of reinforcing metal substructure.

2- Because of the need for a shoulder type margin circumferentially, significant tooth reduction is necessary on the proximal and lingual aspects.

Remember the “Unforgiving” nature of porcelain if an inadequate tooth preparation goes uncorrected, can result in fracture.

3- All-ceramic crowns are not effective as retainers for fixed partial denture especially in posterior region.

Indications:

- High esthetic requirements.
- Considerable proximal or facial caries, that no longer be effectively restored with composite resin especially in anterior teeth.



Contraindications:

1- In molar rarely we use all ceramic crowns due to increased occlusal loads and the reduced esthetic demand make metal-ceramic crown the treatment of choice.

2- Teeth with short clinical crown don't have enough preparation length to support the lingual and incisal surface of the restoration.

3- Thin teeth faciolingually.

4- Bruxism.

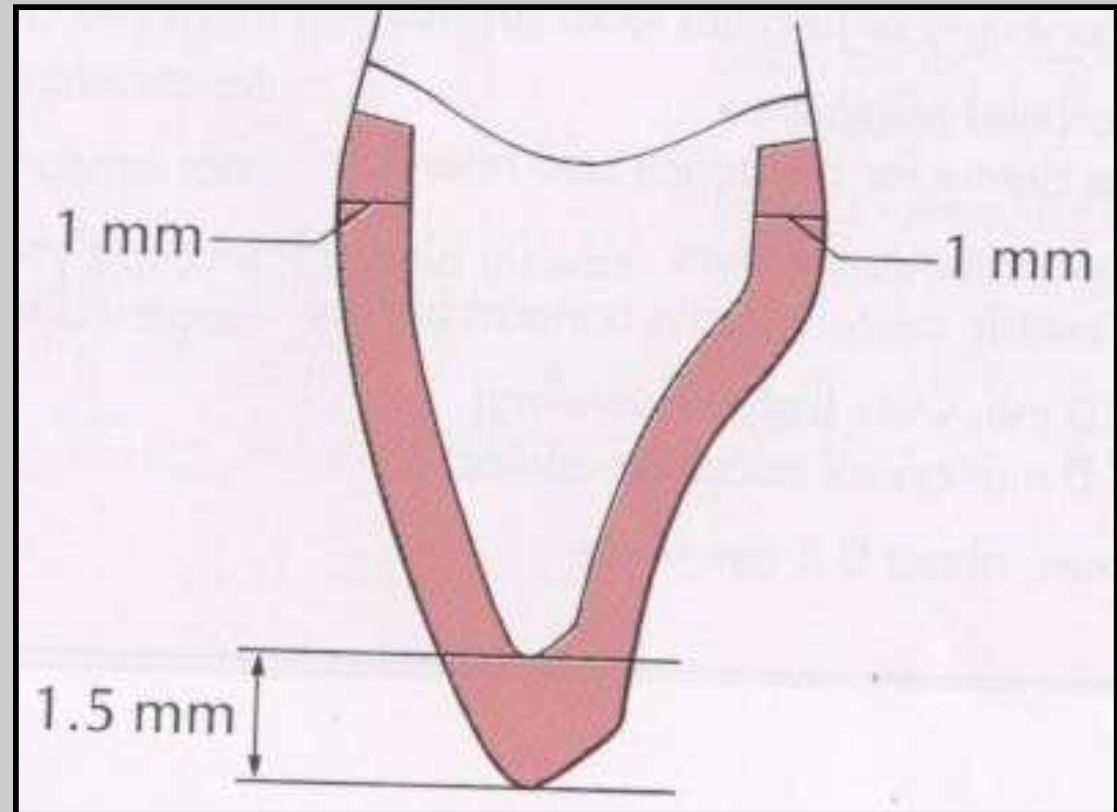
5- Occlusal disturbances like edge-to-edge occlusion that will produce stress in the incisal area of the crowns, also it should not be used when the opposing teeth occludes on the cervical third of the lingual surface , tension will produced half-moon fracture.

Procedure of preparation

The preparation of the tooth for a jacket crown resembles that of the preparation for full metal with facing except that the jacket crown needs:

1-A uniform 1 mm reduction all around the tooth.

2-A butt shoulder finishing line is done all around the tooth.



Steps in preparation:

Prior to preparation a silicone index must be prepared:

Incisal preparation:

Complete reduction of incisal edge should provide 1.5- 2 mm of clearance for porcelain in all mandibular movements. This is important to have cosmetically pleasing restoration with adequate strength. Taper diamond bur is used, placed parallel to the incisal inclination (for posterior teeth 2mm occlusal clearance is needed in all cusps).

1- Depth orientation grooves 1.3 mm in depth are made on the incisal edge using flat-ended taper fissure bur parallel to the incisal inclination of the prepared incisal edge.

2-Any tooth structure between grooves should be removed using the same bur at the same angle.(1.5mm)

3) Incisal clearance then checked in centric and eccentric relations.

Labial (facial) reduction

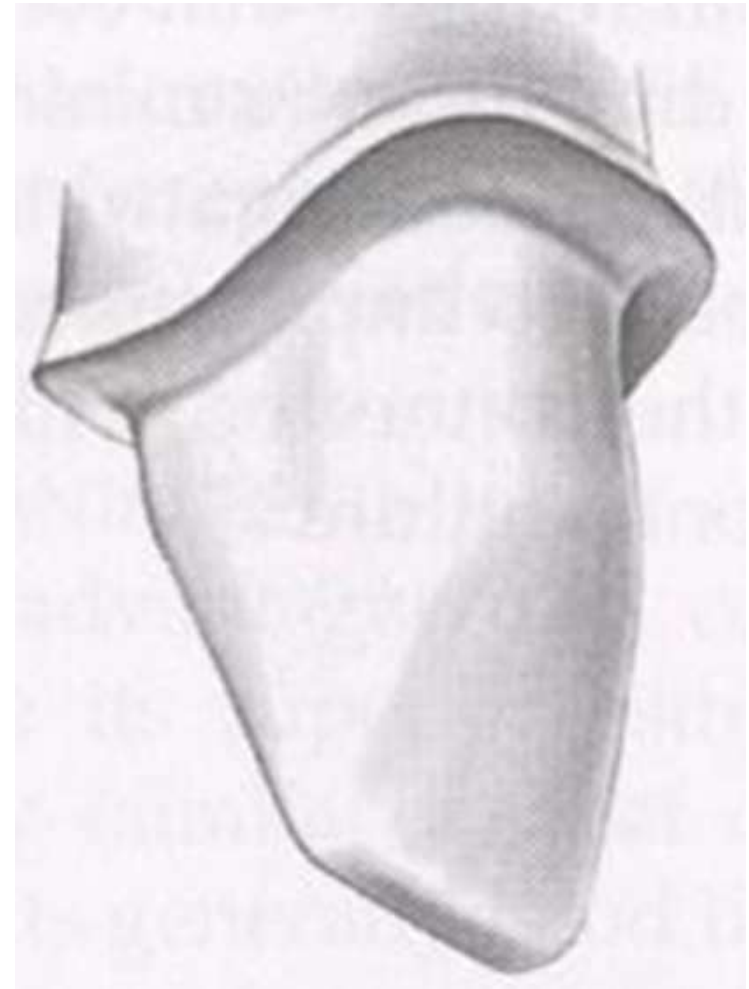
Two plane reduction:

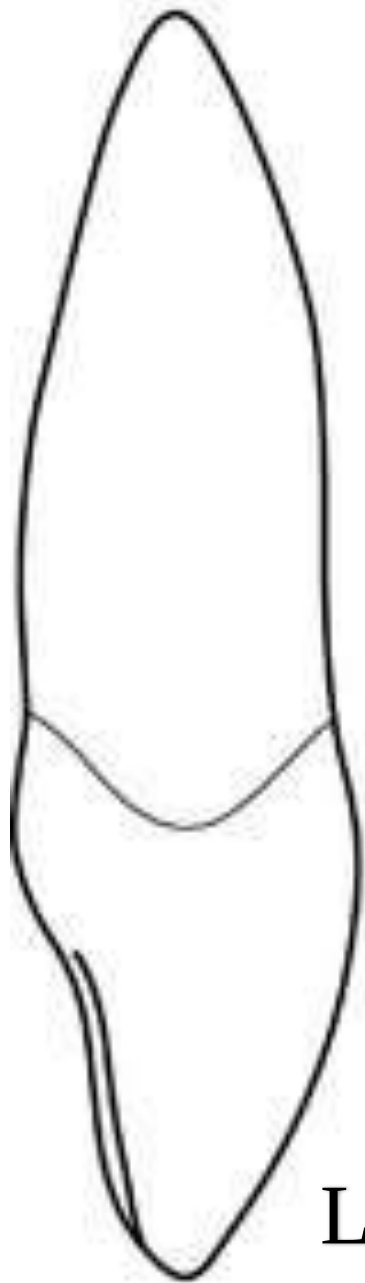
Because of the anatomy of the tooth labially it should be reduced in two planes corresponding to the two geometric planes of the labial surface gingival and incisal planes.

Incisal plane:

1- Three depth orientation grooves 1mm are placed with angle parallel to the inclination of the area.

2- Any tooth structure between grooves should be removed following the contour of the tooth by keeping the bur at the same angle.





Labial and lingual convexities

Labial (facial) reduction

gingival plane:

- 1- 1mm depth grooves placed in gingival part of labial surface parallel to the long axis of the tooth.
- 2- Any tooth structure between grooves should be removed using flat-end taper fissure bur to create shoulder finishing line.



Lingual reduction:

Same preparation used for metal with facing crown but with deeper reduction with 1mm depth.

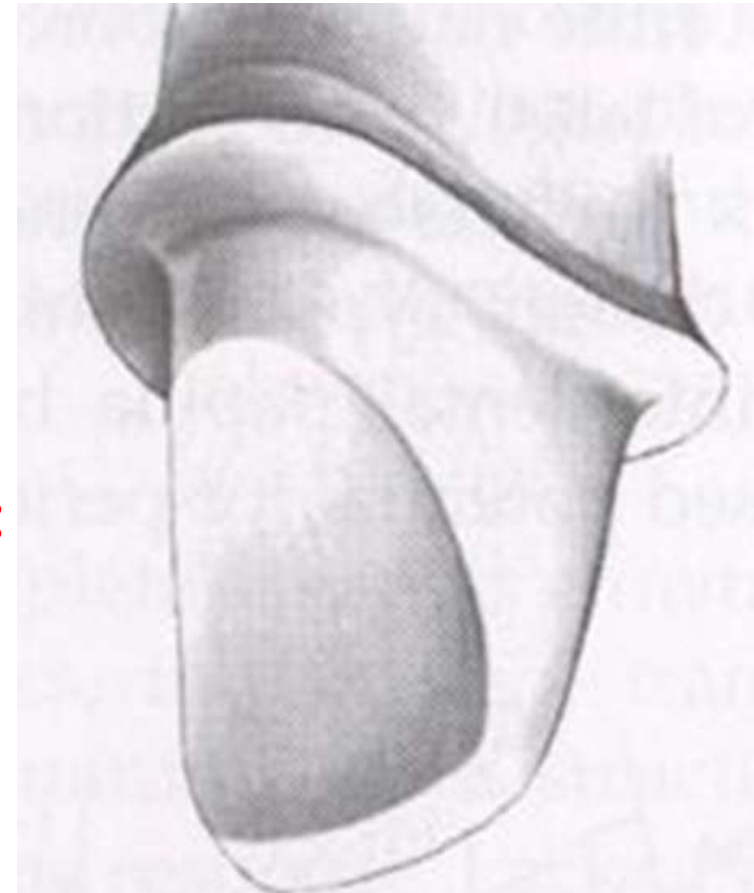
A- Lingual axial reduction:

0.8 mm depth groove placed parallel to the long axis of the tooth.

Flat-end taper fissure bur is used to reduce this area to create shoulder finishing line

B-Cingulum area (fossa) reduction:

0.8mm depth groove placed in the center then small wheel diamond bur is used to reduce this area following its inclination and irregularities.



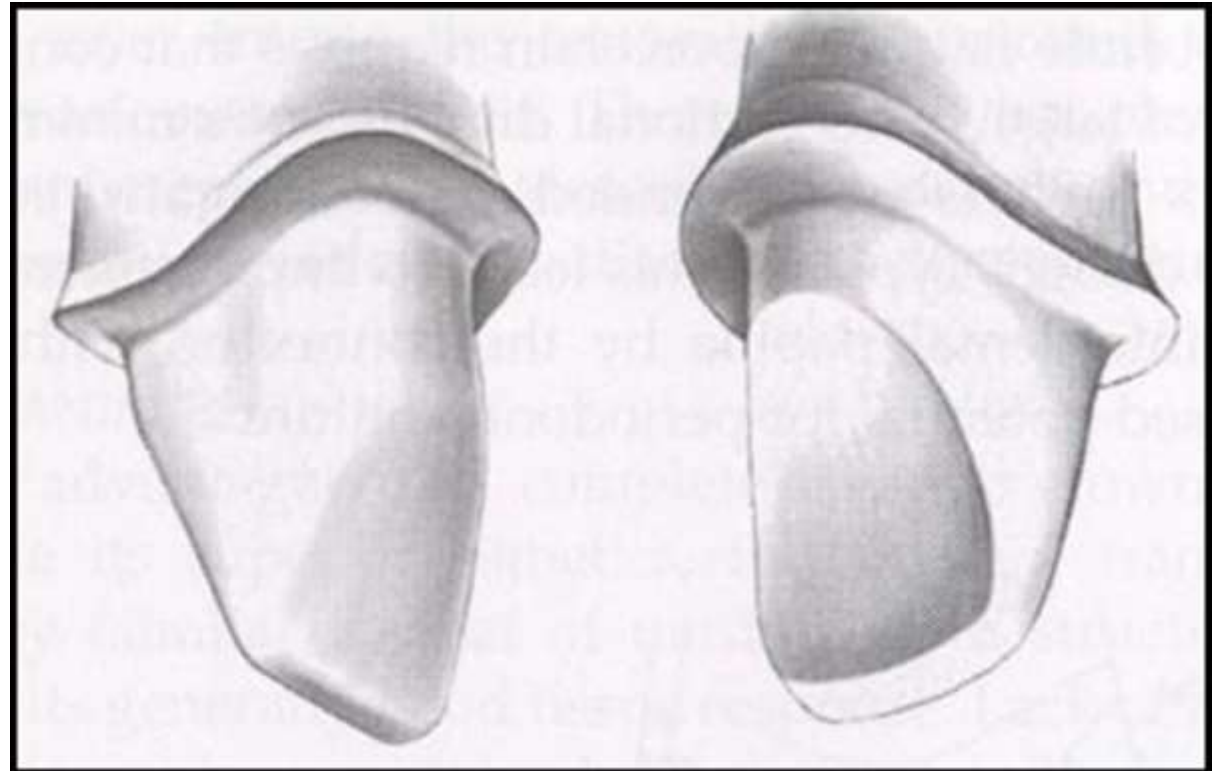
Proximal reduction:

Preparation of proximal surface is done in the same manner as in full-metal with facing except that the shoulder of proximal is continued with the lingual shoulder.

Silicon index can be used to check amount of tooth reduction.

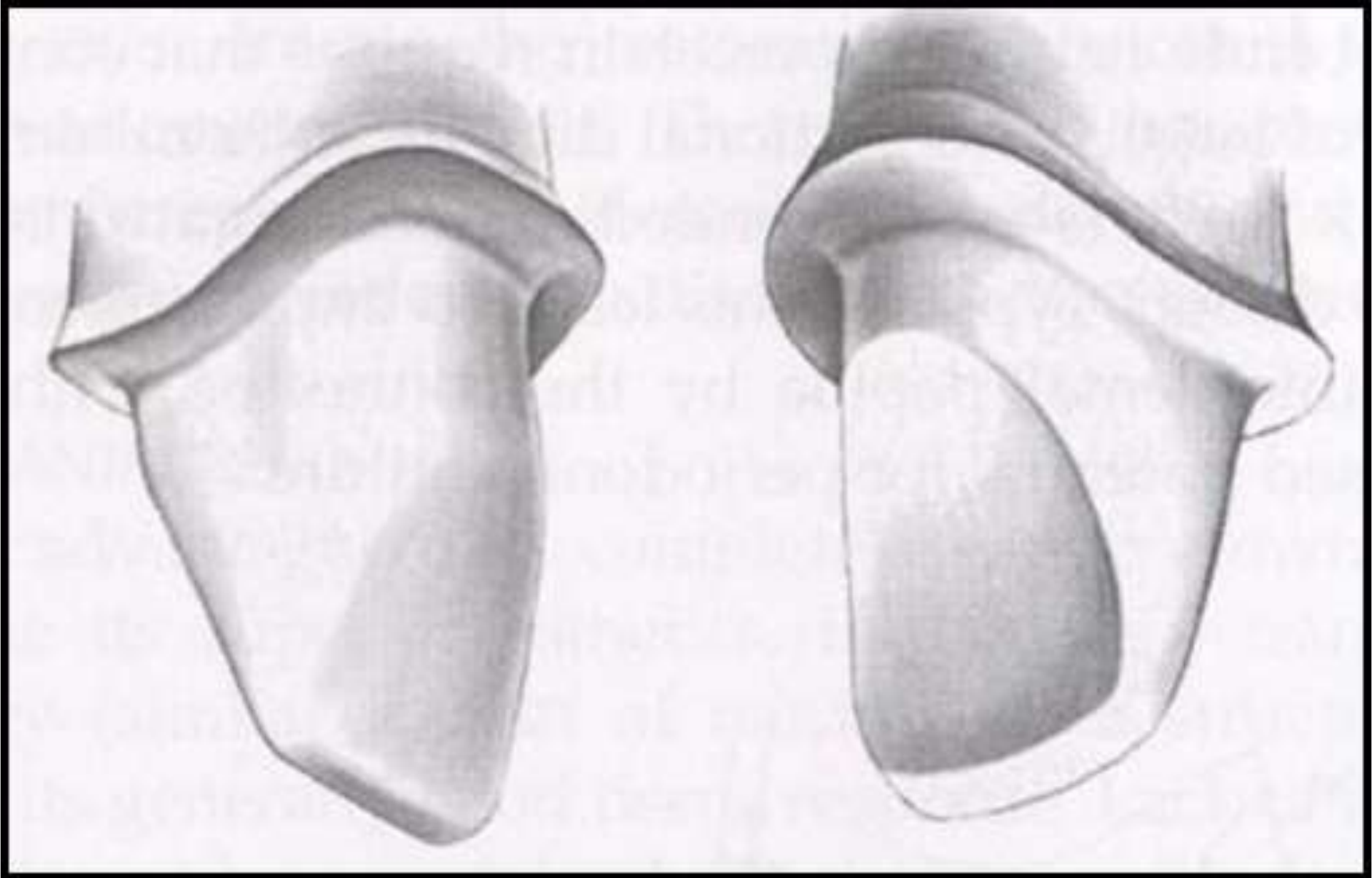
Smoothing of preparation:

Finally smoothing and rounding of all surfaces is done to remove sharp angles and unsupported enamel.



Types of finishing lines used for porcelain jacket crown (all-ceramic)

shoulder or radial shoulder all around has been advocated as gingival finishing lines to be used with porcelain jacket crown preparation. The depth and contour of shoulder is established with the tip of flat-end taper fissure bur. Sharp angles should be rounded to avoid creation of points of stress concentration.



COMPLETE ALL-CERAMIC PREP.

Important points:

1- **To gain intact incisal edge:** Thickness of porcelain should not exceed 2mm, otherwise brittle failure of the material will occur.

2-Favorable distribution of occlusal load: Centric contacts are best confined to the middle third of the lingual surface **but** leaving the crown out of contact is not recommended because future eruption may lead to protrusive interference, which results in fracture of the crown.

Acrylic jacket crown

This is totally made from tooth colored acrylic resin. It can be nearly perfect in appearance at the time of fitting but later on discoloration and loss of contour take place.

Poor adaptation is great disadvantage due to polymerization shrinkage.

It is used in selected cases such as young patients for whom other types of crown restorations are planned but delayed until complete eruption of tooth take place.

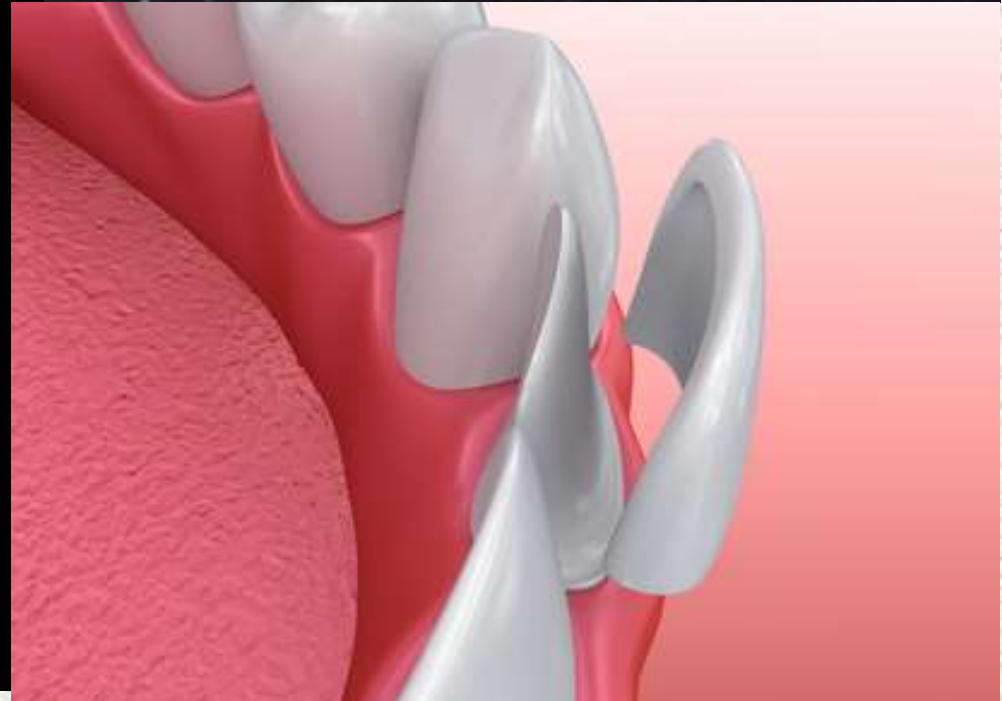
The acrylic jacket crown may be used as a temporary crown or in a case of young patient under 18 years of age, to wait for full eruption of his teeth to allow constructing permanent porcelain jacket crown for him later on.



Disadvantages of acrylic crowns

1. Poor marginal fitness.
2. Poor tissue response.
3. Discoloration with time.
4. Loss of contour (wear easily).

Porcelain laminates veneers



Uses:

1. Improve the color of stained teeth.

2. Alter contours of misshaped teeth.

3. Close interproximal



CERAMIC VENEERS

It consists of thin shell of porcelain applied directly to tooth structure; it is a conservative method of restoring the appearance of discolored, pitted, or fractured anterior teeth. It consists of bonding a thin ceramic laminates onto the labial surfaces of Affected anterior teeth.



Advantages:

The main advantage of facial veneers is that they are conservative to tooth structure. Typically only about 0.5 mm of facial reduction is needed. Since this is confined to the enamel layer, local anesthesia is not usually required.

- Wear and stain resistance.
- Have largely replaced metal-ceramic crowns for the treatment of multiple discolored teeth.

Disadvantage

The main disadvantage is the difficulty in obtaining restoration that is not excessively contoured.

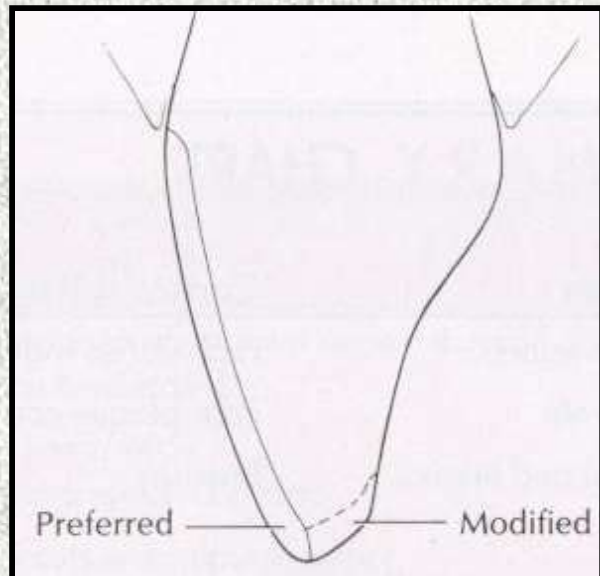
Preparation:-

Preparation is minimal and remaining within enamel

- The gingival third and proximal line angles are often over contoured with these restorations. Therefore, maximum reduction should be achieved with minimum penetration into the dentin.
- Make series of depth holes with a round bur, the required amount of reduction will depend on the extent of discoloration. A minimum of 0.5mm is usually adequate. The reduction should follow the anatomic contours of the tooth.
- Place the (long chamfer) margin. This design has 135° cavosurface angle which exposes the enamel prism ends at the margin for better etching and bonding.
- Wherever possible, place the preparation margin labial to the proximal contact area to preserve it in enamel. Sometimes the proximal margins are extended lingually to include existing composite restoration.



- If possible, do not reduce the incisal edge, this helps support the porcelain and makes chipping less likely.
- If the incisal edge length is to be increased, the preparation should extend to the lingual, care is taken to avoid undercuts with this modification.
- To prevent areas of stress concentration in the porcelain, be sure that all prepared surfaces are rounded.



*Thank
You*



Any question

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